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FTS Infrared Measurements of the Rotational and Vibrational Spectrum of LiH and LiD

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Fourier transform spectra have been measured for the rotational ($\Delta v = 0$) and vibrational ($\Delta v = 1$) spectra of lithium hydride and lithium deuteride in the gas phase. Dunham potential constants and rovibrational constants are given for LiH and LiD. Difficulties were encountered in attempts to fit all four isotopic species to a single modified Dunham potential function. Even data for a single isotopic species could not be fit to a Dunham potential function without significant systematic deviations. These problems are attributed to breakdown in the Born–Oppenheimer approximation.
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INTRODUCTION

Electronic transitions involving the ground electronic state of lithium hydride (LiH) have been extensively studied. The transitions studied cover vibrational states up to $v = 27$. That work has been well summarized by Chan *et al.* (*1*) where references to earlier papers are given.

Because of the accuracy obtainable, the microwave measurements made by Pearson and Gordy (*2*) and by Plummer *et al.* (*3*) have been particularly valuable in defining the rotational constants in the first two vibrational states of lithium hydride and deuteride.

Early infrared measurements of James *et al.* (*4*) were used to estimate some of the terms of the dipole moment function of LiH. Recent infrared measurements of the fundamental and hot bands of lithium hydride by Yamada and Hirota (*5*) have greatly improved the vibrational constants and have also given a value for the transition moment of the fundamental band.

Wharton *et al.* (*6*) have shown that the dipole moment of LiH is very large (5.882 D). With such a large dipole moment the rotational spectrum should be strong and readily observed in the far-infrared region. We have measured the high-J rotational spectrum of ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD with a Fourier transform spectrometer (FTS) in the region from 100 to 280 cm^{-1}.

In this paper we also report, for the same four isotopic species, new measurements of the fundamental bands (and hot bands) made with a tunable diode laser and then with a Fourier transform spectrometer. The present infrared measurements are much

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more extensive than those reported earlier by Yamada and Hirota but are for somewhat higher pressures. The higher pressures raise the possibility of greater pressure shift effects.

EXPERIMENTAL DETAILS

All of the measurements were made with an absorption tube that consisted of a stainless steel heatpipe about 1.55 m long enclosed in a tube oven that heated the central 45 cm of the heatpipe. The ends of the heatpipe were at room temperature and were sealed with NaCl windows for the ordinary infrared region or sealed with polymethylpentene (TPX) windows for the far-infrared region. The tunable diode laser measurements were made with a pipe with a diameter of 2.5 cm while the FTS measurements used a pipe with a diameter of 5 cm.

The heatpipe was loaded with either LiD or LiH, solid materials that are commercially available. For measurements on the ^{6}Li containing species, lithium metal enriched to 95% ^{6}Li was also added to the heatpipe. The heatpipe was connected to a pump and the hydrogen in excess of 25 Torr was removed as it was given off due to the decomposition of the LiD or LiH at high temperatures.

During a typical experiment the heatpipe was heated rapidly to about 700°C and then the temperature increase was slowed and spectral measurements were recorded as the temperature slowly rose to about 800°C, at which point all of the solid LiH had decomposed. The FTS measurements reported here were all taken between 730 and 760°C. During the measurements the pressure of the hydrogen gas was maintained at about 25 Torr (3.3 kPa). No additional hydrogen (or deuterium) gas was used beyond what was generated by the decomposition of the LiH (or LiD).

The far-infrared spectra were calibrated by means of the residual water vapor lines in the absorption path. The wavenumbers given by Johns (*7*) were used. Measurements made on different days were in agreement within 0.0004 cm^{-1}. Both the diode and FTS measurements of the fundamental bands used the ν_3 and $2\nu_2$ bands of both OCS (*8, 9*) and N_2O (*10, 11*) for calibration. In addition some of the diode laser measurements used the 10^00–00^01 band transitions of N_2O near 940 cm^{-1} (*10*). The FTS fundamental band measurements were calibrated by a separate measurement of the OCS spectrum recorded a day after the LiH measurements. In one case a spectrum of N_2O was recorded simultaneously with the LiH spectrum. The uncertainty in the calibration of the FTS measurements of the fundamental band is about 0.001 cm^{-1} while the uncertainty in the far-infrared measurements is estimated to be 0.0004 cm^{-1}. These estimates are based in part on the agreement between measurements made days or weeks apart.

A portion of the far-infrared spectrum of LiD enriched in ^{6}Li is shown in Fig. 1 and a portion of the near-infrared spectrum of LiH enriched in ^{6}Li is shown in Fig. 2.

ASSIGNMENT AND ANALYSIS

The new infrared measurements are given in Tables I, II, and III. They were easily assigned from the line positions calculated from the constants given by Chan *et al.* (*1*). The most recent work of Yamada and Hirota (*5*) confirmed the assignments.

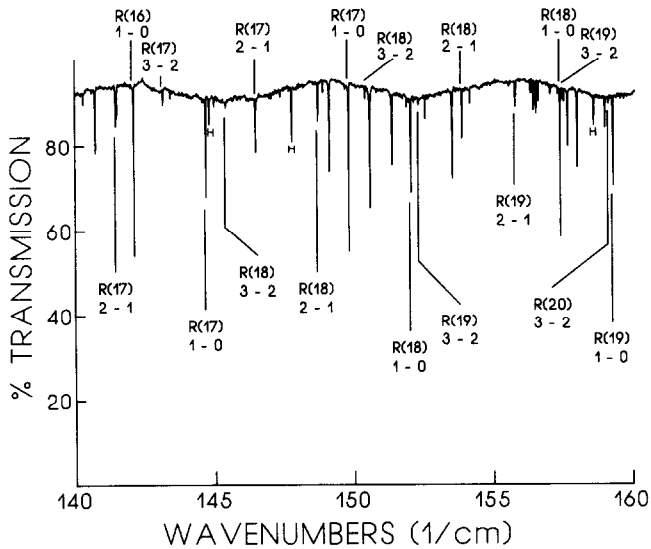


FIG. 1. A portion of the far-infrared spectrum of LiD enriched with ^{6}Li. Lines identified above the spectrum are due to ^{6}LiD and those below the spectrum are due to ^{7}LiD. Lines marked H are due to LiH and some unmarked lines are due to H_2O in the optical path.

As was done in earlier work on TlCl (*12*), the transitions were fit in two different ways. One was a fit to a set of empirical rovibrational constants and the other was a nonlinear fit to the Dunham potential function, which then was used to determine the rovibrational constants. The microwave values of Pearson and Gordy (*2*) and Plummer *et al.* (*3*) were included in the least-squares fits, as were the diode laser measurements of Yamada and Hirota (*5*).

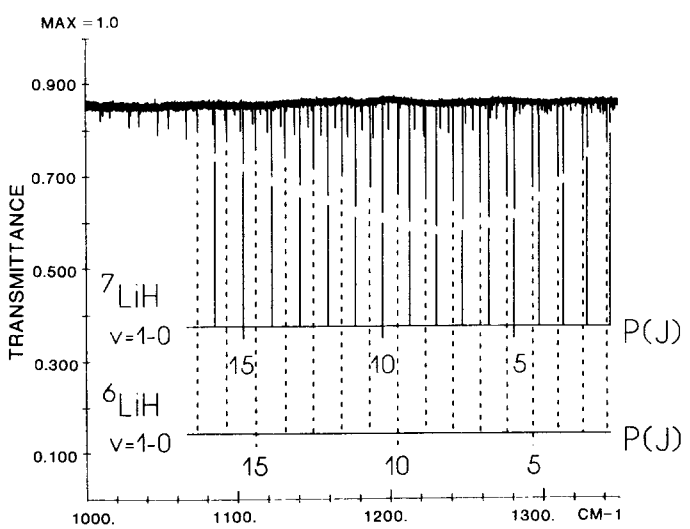


FIG. 2. A portion of the near-infrared spectrum of LiH enriched with ^{6}Li.

TABLE I

Diode Laser Measurements on Lithium Hydride and Deuteride

Transition Rot.	v'-v"	Observed Wavenumber (cm^{-1})	Obs-Calc (cm^{-1})×10^4	Transition Rot.	v'-v"	Observed Wavenumber (cm^{-1})	Obs-Calc (cm^{-1})×10^4
^{7}LiH				^{7}LiD			
P(27)	1 - 0	877.0095	-4	P(21)	1 - 0	825.5346	-1
P(26)	1 - 0	895.9787	18	P(19)	1 - 0	846.6756	3
P(3)	1 - 0	1314.0914	10	P(17)	1 - 0	867.5895	-2
P(2)	1 - 0	1329.6879	7	P(14)	1 - 0	898.4327	16
P(6)	2 - 1	1223.0005	15	P(13)	1 - 0	908.5456	4
R(0)	2 - 1	1328.8194	11	R(14)	1 - 0	1129.6866	3
R(3)	3 - 2	1322.4843	5	P(17)	2 - 1	845.4474	4
^{6}LiD				P(14)	2 - 1	875.6714	2
P(22)	1 - 0	825.0852	10	P(12)	2 - 1	895.4004	11
P(20)	1 - 0	847.1367	10	P(8)	2 - 1	933.6133	-2
P(18)	1 - 0	868.9830	11	P(15)	3 - 2	843.4731	8
P(19)	2 - 1	835.6391	-1	P(11)	3 - 2	882.1201	9
P(16)	2 - 1	867.4671	6				
P(13)	2 - 1	898.6117	35				
P(12)	2 - 1	908.8068	15				
P(15)	3 - 2	854.9814	18				

The measurements were weighted by the inverse squares of their estimated uncertainties. The far-infrared measurements were given an uncertainty of 0.0006 cm^{-1} and the near-infrared FTS measurements were given uncertainties ranging from 0.0008 to 0.0015 cm^{-1} depending on the intensity of the transitions and on how many measurements were averaged.

The fit to empirical rovibrational constants used the form suggested by Watson (*13*). Both the mass invariant constants, U_{ij}, Δ_{ij}^{Li}, and Δ_{ij}^{H}, and the isotopically specific constants, Y_{ij} were determined and are given in Tables IV and V. These constants are related to the observed transitions, ν_{obs}, by the equations

$$G(v, J) = \sum Y_{ij}(v + 1/2)^i[J(J + 1)]^j \tag{1}$$

and

$$\nu_{obs} = G' - G''. \tag{2}$$

The mass invariant constants are related to the Y's by

$$Y_{ij} = U_{ij}[1 + (M_e/M_{Li})(\Delta_{ij}^{Li}) + (M_e/M_H)(\Delta_{ij}^{H})]/\mu^{(i+2j)/2}. \tag{3}$$

Here M_e, M_{Li}, and M_H are the masses of the electron, lithium atom, and hydrogen (or deuterium) atom. The masses were given by Cohen and Taylor (*14*) and by Peiser (*15*). The reduced mass of the molecule is given by $\mu = M_{Li}M_H/(M_{Li} + M_H)$.

Twenty-two rovibrational constants (including the Watson constants) were needed to fit the data for all four isotopes to within experimental error. Initially the fit included the same constants as those used by Yamada and Hirota, but later the constants Y_{40}, Y_{31}, Y_{22}, Y_{13}, and Y_{05} were added to the fit because they were determinable and they gave a better fit. In order to reduce truncation errors and improve the reliability of the constants for extrapolation we also fixed the terms Y_{14} and Y_{06} at the values given by the potential constants. We would have liked to fix the value of Y_{50} also but it was

TABLE II

Far-Infrared Measurements of Rotational Transitions of Lithium Hydride and Deuteride

Transition J"	v"	Observed Wavenumber (cm^{-1})	Obs-Calc (cm^{-1})×10^4
^{6}LiH			
6	0	104.6302	-3
7	0	119.1616	3
8	0	133.5287	-1
9	0	147.7147	1
10	0	161.7012	4
11	0	175.4709	5
12	0	189.0070	1
13	0	202.2953	2
14	0	215.3203	4
15	0	228.0673	-4
17	0	252.6803	-6
18	0	264.5225	-4
19	0	276.0414	3
7	1	115.6878	-18
8	1	129.6335	4
9	1	143.3976	-7
10	1	156.9679	1
11	1	170.3257	8
13	1	196.3392	6
14	1	208.9657	1
15	1	221.3214	5
^{7}LiH			
6	0	102.5281	0
8	0	130.8660	5
9	0	144.7808	-4
10	0	158.5058	6
11	0	172.0210	0
12	0	185.3132	4
13	0	198.3653	-4
14	0	211.1659	7
15	0	223.6994	12
17	0	247.9145	-3
18	0	259.5762	7
19	0	270.9243	1
20	0	281.9522	5
9	1	140.5950	2
10	1	153.9143	1
11	1	167.0302	7
12	1	179.9259	7
13	1	192.5868	3
14	1	205.0000	4
15	1	217.1517	4
16	1	229.0293	1
18	1	251.9186	-2
19	1	262.9099	2
20	1	273.5865	5
21	1	283.9393	2
9	2	136.4822	-1
10	2	149.4044	1
11	2	162.1264	2
12	2	174.6323	-2
13	2	186.9100	10
14	2	198.9431	11
15	2	210.7187	-2
16	2	222.2269	-4
19	2	255.0326	-14
21	2	275.3767	-20
^{6}LiD			
11	0	102.2282	-1
12	0	110.3817	0
13	0	118.4497	0
14	0	126.4266	1
15	0	134.3065	-4
16	0	142.0850	-4
17	0	149.7565	-4
18	0	157.3160	-5
19	0	164.7591	-4
20	0	172.0814	-1
21	0	179.2780	0
22	0	186.3452	0
23	0	193.2790	-2
24	0	200.0756	-8
25	0	206.7326	-6
26	0	213.2467	1
27	0	219.6140	4
28	0	225.8322	7
12	1	107.9399	-11

Transition J"	v"	Observed Wavenumber (cm^{-1})	Obs-Calc (cm^{-1})×10^4
^{6}LiD			
13	1	115.8274	0
14	1	123.6234	-6
15	1	131.3250	-4
16	1	138.9266	2
17	1	146.4224	2
18	1	153.8073	-4
19	1	161.0781	-3
20	1	168.2296	-1
21	1	175.2577	0
22	1	182.1574	-8
23	1	188.9274	-1
25	1	202.0583	-3
26	1	208.4146	8
27	1	214.6250	0
28	1	220.6914	21
12	2	105.5339	-1
13	2	113.2410	-2
14	2	120.8585	-14
15	2	128.3850	1
16	2	135.8104	-6
17	2	143.1330	-3
18	2	150.3458	-12
19	2	157.4473	-2
20	2	164.4301	-5
21	2	171.2925	4
22	2	178.0276	-6
23	2	184.6331	-19
24	2	191.1098	8
25	2	197.4492	20
26	2	203.6463	-1
^{7}LiD			
12	0	106.5383	0
13	0	114.3400	1
14	0	122.0556	-8
15	0	129.6830	0
16	0	137.2139	-5
17	0	144.6454	-6
18	0	151.9733	0
19	0	159.1917	2
20	0	166.2964	-3
22	0	180.1512	-4
23	0	186.8935	-3
24	0	193.5075	-4
25	0	199.9898	-8
26	0	206.3388	-2
28	0	218.6213	3
29	0	224.5494	-3
30	0	230.3346	10
12	1	104.2262	4
13	1	111.8534	-14
14	1	119.4008	5
15	1	126.8563	-7
16	1	134.2202	0
17	1	141.4842	-7
19	1	155.7008	-3
20	1	162.6442	2
21	1	169.4716	0
22	1	176.1797	-2
23	1	182.7653	-1
24	1	189.2245	-1
26	1	201.7523	3
27	1	207.8150	6
28	1	213.7376	-12
29	1	219.5222	-10
12	2	101.9457	13
15	2	124.0682	-12
16	2	131.2645	-18
17	2	138.3661	-2
18	2	145.3646	-1
19	2	152.2571	-4
20	2	159.0403	-2
21	2	165.7087	-9
22	2	172.2625	12
23	2	178.6928	7
24	2	185.0008	22
25	2	191.1790	15
27	2	203.1412	-4

TABLE III

Infrared Transitions[a] of Lithium Hydride and Deuteride

Transition Rot.	v'-v"	Observed Wavenumber	o-c	Transition Rot.	v'-v"	Observed Wavenumber	o-c	Transition Rot.	v'-v"	Observed Wavenumber	o-c
^{6}LiH				^{7}LiH				R(12)	2 - 1	1457.3024	-1
P(20)	1 - 0	1016.4000	8	P(20)	1 - 0	1010.2730	12	R(13)	2 - 1	1464.2893	30
P(19)	1 - 0	1035.8050	-3	P(19)	1 - 0	1029.2776	4	P(14)	3 - 2	1047.1536	2
P(18)	1 - 0	1055.1608	-3	P(18)	1 - 0	1048.2312	4	P(13)	3 - 2	1064.6358	10
P(17)	1 - 0	1074.4497	-3	P(17)	1 - 0	1067.1163	0	P(12)	3 - 2	1081.9629	0
P(16)	1 - 0	1093.6539	-6	P(16)	1 - 0	1085.9163	-8	P(11)	3 - 2	1099.1182	-12
P(15)	1 - 0	1112.7563	-3	P(15)	1 - 0	1104.6147	-10	P(10)	3 - 2	1116.0837	-16
P(14)	1 - 0	1131.7373	-7	P(14)	1 - 0	1123.1931	-13	P(9)	3 - 2	1132.8398	-16
P(13)	1 - 0	1150.5800	5	P(13)	1 - 0	1141.6335	-13	P(8)	3 - 2	1149.3671	-9
P(12)	1 - 0	1169.2615	0	P(12)	1 - 0	1159.9168	-13	P(7)	3 - 2	1165.6453	-1
P(11)	1 - 0	1187.7640	0	P(11)	1 - 0	1178.0233	-14	P(6)	3 - 2	1181.6525	-9
P(10)	1 - 0	1206.0660	-5	P(10)	1 - 0	1195.9338	-14	P(5)	3 - 2	1197.3724	5
P(9)	1 - 0	1224.1477	-3	P(9)	1 - 0	1213.6281	-12	R(3)	3 - 2	1322.4854	16
P(8)	1 - 0	1241.9874	1	P(8)	1 - 0	1231.0855	-11	R(4)	3 - 2	1334.2110	0
P(7)	1 - 0	1259.5625	-2	P(7)	1 - 0	1248.2853	-10	R(5)	3 - 2	1345.4357	-9
P(6)	1 - 0	1276.8521	-6	P(6)	1 - 0	1265.2067	-10	R(8)	3 - 2	1375.9370	-62
P(5)	1 - 0	1293.8356	5	P(5)	1 - 0	1281.8286	-10	^{6}LiD			
P(4)	1 - 0	1310.4888	8	P(4)	1 - 0	1298.1301	-8	P(25)	1 - 0	791.7258	-10
P(3)	1 - 0	1326.7895	1	P(3)	1 - 0	1314.0895	-9	P(24)	1 - 0	802.8740	-16
P(2)	1 - 0	1342.7186	13	P(2)	1 - 0	1329.6865	-7	P(23)	1 - 0	813.9978	11
P(1)	1 - 0	1358.2492	-7	P(1)	1 - 0	1344.8996	-7	P(22)	1 - 0	825.0844	2
R(1)	1 - 0	1402.2610	-12	R(0)	1 - 0	1374.0919	-10	P(21)	1 - 0	836.1314	-12
R(2)	1 - 0	1416.0008	-8	R(1)	1 - 0	1388.0316	-4	P(20)	1 - 0	847.1345	-12
R(3)	1 - 0	1429.2431	13	R(2)	1 - 0	1401.5059	-7	P(19)	1 - 0	858.0877	1
R(4)	1 - 0	1441.9637	1	R(3)	1 - 0	1414.4974	-3	P(18)	1 - 0	868.9822	3
R(5)	1 - 0	1454.1493	7	R(4)	1 - 0	1426.9863	-8	P(17)	1 - 0	879.8116	-6
R(6)	1 - 0	1465.7791	0	R(5)	1 - 0	1438.9560	-9	P(16)	1 - 0	890.5718	-4
R(7)	1 - 0	1476.8369	-12	R(6)	1 - 0	1450.3897	-2	P(15)	1 - 0	901.2551	-1
R(8)	1 - 0	1487.3109	10	R(7)	1 - 0	1461.2696	-5	P(14)	1 - 0	911.8541	-3
R(9)	1 - 0	1497.1816	22	R(8)	1 - 0	1471.5815	-5	P(13)	1 - 0	922.3628	-2
R(10)	1 - 0	1506.4339	12	R(9)	1 - 0	1481.3110	-2	P(12)	1 - 0	932.7741	-1
R(11)	1 - 0	1515.0569	1	R(10)	1 - 0	1490.4436	-6	P(11)	1 - 0	943.0807	-1
R(12)	1 - 0	1523.0405	6	R(11)	1 - 0	1498.9679	-6	P(10)	1 - 0	953.2760	2
R(13)	1 - 0	1530.3714	-1	R(12)	1 - 0	1506.8721	-7	P(9)	1 - 0	963.3520	-1
P(19)	2 - 1	999.2107	4	R(13)	1 - 0	1514.1453	-12	P(8)	1 - 0	973.3026	3
P(17)	2 - 1	1036.8416	-10	R(14)	1 - 0	1520.7804	-1	P(7)	1 - 0	983.1203	7
P(16)	2 - 1	1055.5420	6	R(15)	1 - 0	1526.7676	10	P(6)	1 - 0	992.7967	3
P(15)	2 - 1	1074.1390	3	P(18)	2 - 1	1011.7880	21	P(5)	1 - 0	1002.3254	-1
P(14)	2 - 1	1092.6178	12	P(17)	2 - 1	1030.1823	17	P(4)	1 - 0	1011.6998	1
P(13)	2 - 1	1110.9560	-4	P(16)	2 - 1	1048.4914	5	P(3)	1 - 0	1020.9118	0
P(12)	2 - 1	1129.1388	-2	P(15)	2 - 1	1066.6999	-2	P(2)	1 - 0	1029.9551	5
P(11)	2 - 1	1147.1444	-3	P(14)	2 - 1	1084.7904	-3	P(1)	1 - 0	1038.8208	-3
P(10)	2 - 1	1164.9534	0	P(13)	2 - 1	1102.7452	4	R(0)	1 - 0	1055.9972	8
P(9)	2 - 1	1182.5453	4	P(12)	2 - 1	1120.5429	-10	R(1)	1 - 0	1064.2925	10
P(8)	2 - 1	1199.8983	3	P(11)	2 - 1	1138.1682	-9	R(2)	1 - 0	1072.3836	12
P(7)	2 - 1	1216.9916	-3	P(10)	2 - 1	1155.6005	-6	R(3)	1 - 0	1080.2625	-1
P(6)	2 - 1	1233.8051	-1	P(9)	2 - 1	1172.8197	-6	R(4)	1 - 0	1087.9254	0
P(5)	2 - 1	1250.3156	-7	P(8)	2 - 1	1189.8058	-7	R(5)	1 - 0	1095.3645	-2
P(4)	2 - 1	1266.5036	-1	P(7)	2 - 1	1206.5390	-6	R(6)	1 - 0	1102.5746	3
P(3)	2 - 1	1282.3450	-8	P(6)	2 - 1	1222.9980	-10	R(7)	1 - 0	1109.5480	0
P(2)	2 - 1	1297.8211	0	P(5)	2 - 1	1239.1636	-5	R(8)	1 - 0	1116.2800	-2
R(1)	2 - 1	1355.6288	-13	P(4)	2 - 1	1255.0137	-5	R(9)	1 - 0	1122.7651	-2
R(2)	2 - 1	1368.9536	-25	P(3)	2 - 1	1270.5279	-7	R(10)	1 - 0	1128.9983	4
R(4)	2 - 1	1394.1182	-5	P(2)	2 - 1	1285.6857	-10	R(11)	1 - 0	1134.9720	-10
R(5)	2 - 1	1405.9182	-2	P(1)	2 - 1	1300.4691	13	R(12)	1 - 0	1140.6859	2
R(6)	2 - 1	1417.1733	-2	R(0)	2 - 1	1328.8186	4	R(13)	1 - 0	1146.1314	0
R(7)	2 - 1	1427.8660	-15	R(1)	2 - 1	1342.3474	-6	R(14)	1 - 0	1151.3066	9
R(8)	2 - 1	1437.9885	37	R(2)	2 - 1	1355.4215	0	R(15)	1 - 0	1156.2058	13
R(9)	2 - 1	1447.5100	-9	R(3)	2 - 1	1368.0198	-5	R(16)	1 - 0	1160.8226	-16
P(14)	3 - 2	1054.2797	-27	R(4)	2 - 1	1380.1257	-5	R(17)	1 - 0	1165.1604	-6
P(13)	3 - 2	1072.1369	22	R(5)	2 - 1	1391.7217	-1	R(19)	1 - 0	1172.9726	-10
P(12)	3 - 2	1089.8318	2	R(6)	2 - 1	1402.7897	-5	P(23)	2 - 1	792.4317	-8
P(10)	3 - 2	1124.6812	-6	R(7)	2 - 1	1413.3153	-4	P(22)	2 - 1	803.2980	-5
P(7)	3 - 2	1175.2988	4	R(8)	2 - 1	1423.2820	-11	P(21)	2 - 1	814.1254	-2
P(6)	3 - 2	1191.6459	0	R(9)	2 - 1	1432.6768	-13	P(20)	2 - 1	824.9075	-3
P(5)	3 - 2	1207.6997	35	R(10)	2 - 1	1441.4880	3	P(19)	2 - 1	835.6392	0
P(4)	3 - 2	1223.4259	-22	R(11)	2 - 1	1449.7019	23	P(18)	2 - 1	846.3136	-1

a) The wavenumbers are given in units of cm^{-1} and the o-c column (obs-calc) is in units of $cm^{-1} \times 10^4$.

TABLE III—*Continued*

Transition Rot.	v'-v"	Observed Wavenumber	o-c	Transition Rot.	v'-v"	Observed Wavenumber	o-c	Transition Rot.	v'-v"	Observed Wavenumber	o-c
P(17)	2 - 1	856.9243	-6	P(18)	1 - 0	857.1627	-10	P(6)	2 - 1	951.9948	2
P(16)	2 - 1	867.4659	-7	P(17)	1 - 0	867.5888	-8	P(5)	2 - 1	960.9804	-1
P(15)	2 - 1	877.9321	0	P(16)	1 - 0	877.9460	-10	P(4)	2 - 1	969.8211	1
P(14)	2 - 1	888.3148	-1	P(15)	1 - 0	888.2290	-6	P(3)	2 - 1	978.5091	-6
P(13)	2 - 1	898.6079	-3	P(14)	1 - 0	898.4305	-6	P(2)	2 - 1	987.0392	-6
P(12)	2 - 1	908.8050	-3	P(13)	1 - 0	908.5445	-7	P(1)	2 - 1	995.4048	-1
P(11)	2 - 1	918.8994	3	P(12)	1 - 0	918.5647	-7	R(1)	2 - 1	1019.4447	-5
P(10)	2 - 1	928.8821	-9	P(11)	1 - 0	928.4846	-5	R(2)	2 - 1	1027.0864	2
P(9)	2 - 1	938.7495	-1	P(10)	1 - 0	938.2971	-7	R(3)	2 - 1	1034.5298	-7
P(8)	2 - 1	948.4928	8	P(9)	1 - 0	947.9965	-4	R(4)	2 - 1	1041.7722	-1
P(7)	2 - 1	958.1037	6	P(8)	1 - 0	957.5750	-5	R(5)	2 - 1	1048.8056	-1
P(6)	2 - 1	967.5758	1	P(7)	1 - 0	967.0264	-8	R(6)	2 - 1	1055.6251	0
P(5)	2 - 1	976.9021	-5	P(6)	1 - 0	976.3445	-4	R(7)	2 - 1	1062.2240	-9
P(4)	2 - 1	986.0763	-6	P(5)	1 - 0	985.5215	-5	R(8)	2 - 1	1068.5993	-7
P(3)	2 - 1	995.0899	-13	P(4)	1 - 0	994.5515	-4	R(9)	2 - 1	1074.7448	-1
P(2)	2 - 1	1003.9396	11	P(3)	1 - 0	1003.4272	-5	R(10)	2 - 1	1080.6546	-5
P(1)	2 - 1	1012.6140	20	P(2)	1 - 0	1012.1424	-5	R(11)	2 - 1	1086.3240	-16
R(0)	2 - 1	1029.4100	10	P(1)	1 - 0	1020.6911	4	R(12)	2 - 1	1091.7515	-3
R(1)	2 - 1	1037.5211	22	R(0)	1 - 0	1037.2571	-10	R(13)	2 - 1	1096.9285	-12
R(2)	2 - 1	1045.4261	-15	R(1)	1 - 0	1045.2639	-9	R(15)	2 - 1	1106.5240	-2
R(3)	2 - 1	1053.1289	5	R(2)	1 - 0	1053.0785	0	P(20)	3 - 2	793.4242	-1
R(4)	2 - 1	1060.6153	4	R(3)	1 - 0	1060.6925	-3	P(18)	3 - 2	813.6324	-9
R(5)	2 - 1	1067.8800	-10	R(4)	1 - 0	1068.1013	-6	P(17)	3 - 2	823.6482	-6
R(6)	2 - 1	1074.9202	-2	R(5)	1 - 0	1075.2994	-3	P(16)	3 - 2	833.5977	6
R(7)	2 - 1	1081.7270	-4	R(6)	1 - 0	1082.2806	1	P(15)	3 - 2	843.4726	3
R(8)	2 - 1	1088.2958	-4	R(7)	1 - 0	1089.0382	-4	P(14)	3 - 2	853.2687	3
R(9)	2 - 1	1094.6216	3	R(8)	1 - 0	1095.5687	-2	P(13)	3 - 2	862.9790	0
R(11)	2 - 1	1106.5213	16	R(9)	1 - 0	1101.8651	-10	P(12)	3 - 2	872.5979	-1
R(12)	2 - 1	1112.0815	-17	R(10)	1 - 0	1107.9243	-6	P(11)	3 - 2	882.1196	5
R(13)	2 - 1	1117.3822	-11	R(11)	1 - 0	1113.7405	-4	P(10)	3 - 2	891.5353	-6
R(14)	2 - 1	1122.4147	-12	R(12)	1 - 0	1119.3084	-8	P(9)	3 - 2	900.8407	-12
P(18)	3 - 2	824.0012	-13	R(13)	1 - 0	1124.6254	-4	P(8)	3 - 2	910.0303	-3
P(17)	3 - 2	834.3991	-2	R(14)	1 - 0	1129.6863	0	P(7)	3 - 2	919.0961	7
P(15)	3 - 2	854.9792	-5	R(15)	1 - 0	1134.4856	-14	P(6)	3 - 2	928.0309	10
P(14)	3 - 2	865.1504	0	R(16)	1 - 0	1139.0241	-1	P(5)	3 - 2	936.8273	-2
P(13)	3 - 2	875.2327	0	P(24)	2 - 1	772.8195	-2	P(4)	3 - 2	945.4818	0
P(12)	3 - 2	885.2185	-14	P(23)	2 - 1	783.3229	-8	P(3)	3 - 2	953.9843	-18
P(11)	3 - 2	895.1070	18	P(22)	2 - 1	793.7937	-4	P(2)	3 - 2	962.3328	-10
P(10)	3 - 2	904.8819	2	P(20)	2 - 1	814.6106	-11	R(2)	3 - 2	1001.5036	10
P(9)	3 - 2	914.5418	-7	P(19)	2 - 1	824.9476	-5	R(3)	3 - 2	1008.7776	-12
P(8)	3 - 2	924.0798	-9	P(18)	2 - 1	835.2277	-7	R(4)	3 - 2	1015.8556	4
P(7)	3 - 2	933.4898	6	P(17)	2 - 1	845.4462	-7	R(5)	3 - 2	1022.7252	-7
P(6)	3 - 2	942.7623	13	P(16)	2 - 1	855.5968	-9	R(6)	3 - 2	1029.3848	-6
P(5)	3 - 2	951.8897	6	P(15)	2 - 1	865.6734	-10	R(7)	3 - 2	1035.8280	-4
γ_{LiD}				P(14)	2 - 1	875.6699	-13	R(8)	3 - 2	1042.0494	0
P(26)	1 - 0	771.9906	-8	P(13)	2 - 1	885.5807	-9	R(9)	3 - 2	1048.0429	-5
P(25)	1 - 0	782.7574	0	P(12)	2 - 1	895.3991	-2	R(10)	3 - 2	1053.8059	2
P(24)	1 - 0	793.4996	-4	P(11)	2 - 1	905.1172	-7	R(11)	3 - 2	1059.3312	-3
P(23)	1 - 0	804.2144	5	P(10)	2 - 1	914.7300	-8	R(12)	3 - 2	1064.6165	2
P(22)	1 - 0	814.8931	-9	P(9)	2 - 1	924.2312	-3	P(10)	4 - 3	868.6908	-11
P(21)	1 - 0	825.5334	-12	P(8)	2 - 1	933.6131	-4	P(9)	4 - 3	877.8055	-10
P(20)	1 - 0	836.1307	4	P(7)	2 - 1	942.8699	-2	P(6)	4 - 3	904.4262	-28
P(19)	1 - 0	846.6745	-8								

not well determined by the potential constants since it depends on a_7 and a_8, which were poorly determined.

Table V gives the empirical rovibrational constants determined by the fit. The column of observed − calculated values given in Tables I, II, and III are based on the transition frequencies calculated by using the constants given in Table V.

The measurements were also fit directly to a Dunham potential function. The basic equations for this fit were given by Dunham (*16*) and later extended by others (*17–20*).

As a first approximation to the correction terms given by Watson (see Eq. (3) above), we have introduced mass invariant terms, U_B and U_ω, for B_e and ω_e as follows:

TABLE IV

Reduced Dunham Rovibrational Constants (in cm^{-1}) and Dimensionless Adiabatic Correction Factors for Lithium Hydride

ij	U_{ij}	Δ^{H}_{ij}	Δ^{Li}_{ij}
10	1319.948 75(267)[a]	-0.721 75(209)	-0.177 22(1279)
20	-20.431 87(212)	-0.477 9(383)	
30	0.145 899(797)		
40	-0.229 99(951)x10^{-2}		
01	6.627 104 73(923)	-1.563 409(1114)	-0.117 22(1502)
11	-0.179 142 52(691)	-0.760 6(409)	
21	0.162 036(501)x10^{-2}		
31	-0.321 26(799)x10^{-4}		
02	-0.668 148 8(690)x10^{-3}	-3.428 8(529)	
12	0.117 000(150)x10^{-4}		
22	-0.131 33(577)x10^{-6}		
03	0.742 74(334)x10^{-7}		
13	-0.558 3(212)x10^{-9}		
04	-0.107 52(598)x10^{-10}		
14	[-0.101][b]x10^{-12}		
05	0.171 1(361)x10^{-14}		
06	[-0.3378]x10^{-18}		
	rms dev. of 574 I.R. measurements	0.00096 cm^{-1}	

a) The uncertainties (twice the standard error) in the last digits are given in parentheses.

b) Values given in square brackets were calculated from the potential constants given in Table VI.

TABLE V

Dunham Rovibrational Constants (in cm^{-1}) for Lithium Hydride

Constant	^{7}LiD	^{6}LiD	^{7}LiH	^{6}LiH
Y_{10}	1054.936 84(179)[a]	1074.331 99(182)	1405.509 36(266)	1420.117 54(270)
Y_{20}	-13.054 89(139)	-13.539 39(144)	-23.179 38(253)	-23.663 82(259)
Y_{30}	0.074 531(402)	0.078 718(425)	0.176 365(952)	0.181 923(982)
$Y_{40}x10^{2}$	-0.093 92(388)	-0.101 02(418)	-0.296 16(1225)	-0.308 67(1276)
Y_{01}	4.233 079 91(176)	4.390 175 70(174)	7.513 751 04(390)	7.670 775 00(381)
Y_{11}	-0.091 493 67(367)	-0.096 634 06(388)	-0.216 460 60(962)	-0.223 281 90(993)
$Y_{21}x10^{2}$	0.066 169(204)	0.071 171(220)	0.208 653(645)	0.217 465(672)
$Y_{31}x10^{4}$	-0.104 87(261)	-0.114 88(286)	-0.440 69(1096)	-0.464 07(1154)
$Y_{02}x10^{3}$	-0.272 590 4(278)	-0.293 199 2(299)	-0.858 766 8(883)	-0.895 038 0(920)
$Y_{12}x10^{4}$	0.038 193 5(489)	0.041 836 5(536)	0.160 492(205)	0.169 009(216)
$Y_{22}x10^{6}$	-0.034 271(1505)	-0.038 230(1679)	-0.191 90(843)	-0.204 19(897)
$Y_{03}x10^{6}$	0.019 382 2(871)	0.021 621 3(971)	0.108 532(488)	0.115 480(519)
$Y_{13}x10^{10}$	-0.116 45(441)	-0.132 30(502)	-0.869 0(329)	-0.934 2(354)
$Y_{04}x10^{12}$	-0.017 930(997)	-0.020 744(1154)	-0.178 29(992)	-0.193 67(1077)
$Y_{14}x10^{14}$	[-0.013 5][b]	[-0.015 9]	[-0.178]	[-0.196]
$Y_{05}x10^{14}$	0.018 23(385)	0.021 87(462)	0.321 9(680)	0.357 0(754)
$Y_{06}x10^{18}$	[-0.023]	[-0.029]	[-0.72]	[-0.82]

a) The uncertainties (twice the standard error) in the last digits are given in parentheses.

b) The values given in square brackets were calculated from the potential constants given in Table VI.

$$B_e = U_B[1 + (M_e/M_{Li})(\Delta_B^{Li}) + (M_e/M_H)(\Delta_B^H)]/\mu \quad (4)$$

and

$$\omega_e = U_\omega[1 + (M_e/M_{Li})(\Delta_\omega^{Li}) + (M_e/M_H)(\Delta_\omega^H)]/\mu^{1/2}. \quad (5)$$

These have worked quite well for heavier molecules but are inadequate for LiH. A similar mass dependence for the Dunham a's was tried but, although seemingly well defined, they were unable to fit the data for all four isotopic species to within experimental error.

When all four isotopic species were included in the potential function fit, the standard deviation was more than four times larger than the fit to the rovibrational constants given in Table IV. Even when only the data for a single isotopic species were fit, the standard deviation was twice as large when fit to a Dunham potential function as when fit to empirical rovibrational constants. In addition, the fits to a potential function showed systematic deviations as large as 0.006 cm^{-1}.

In spite of the poor fit of the data to a Dunham potential function, the Dunham potential constants were used to estimate the values of Y_{14} and Y_{06} which were used in the fit of the other rovibrational constants. Y_{14} and Y_{06} only depend on the values of a_1, a_2, a_3, and a_4 (in addition to B_e and ω_e) which are well determined. The constants resulting from the fit of the data to a Dunham potential function are given in Table VI.

DISCUSSION

In recent studies involving light molecules we have encountered difficulties in fitting spectra to a Dunham potential function, while no similar difficulties were encountered

TABLE VI

Dunham Potential Constants for Lithium Hydride

U ($cm^{-1}u^{k}$)	1319.986 77(1132)[a]
ω_e (cm^{-1})	1055.189 28(905)[b]
Δ_ω^{Li}	-0.163 9(665)
Δ_ω^H	-0.746 42(354)
U_B ($cm^{-1}u$)	6.627 152 8(473)
B_e (cm^{-1})	4.234 952 8(302)
Δ_B^{Li}	-0.136 4(801)
Δ_B^H	-1.578 18(397)
a_1	-1.897 234 4(1284)
a_2	2.441 780(1110)
a_3	-2.633 43(543)
a_4	2.584 2(176)
a_5	-2.669 5(1237)
a_6	3.433(472)
a_7	-4.553(956)
Infrared Transitions:	
number	574
RMS dev. (cm^{-1})	0.00453

a) The uncertainties (twice the standard error) in the last digits are given in parentheses.
b) The values for ω_e and B_e are given for the ^{7}LiD species.

in fitting the same data to rovibrational constants that were not constrained to satisfy a single potential function. These difficulties were only observed because the data were quite accurate; better than ± 0.002 cm^{-1} for infrared data and better than ± 0.2 MHz (± 0.000007 cm^{-1}) for heterodyne measurements of pure rotational transitions. The earlier difficulties involved light molecules, NaH (*21*) and CO (*22*). Although some data on isotopically substituted species were available, the difficulties were found when data were fit for *only one* isotopic species. The present measurements on LiH and LiD fit this same pattern.

In the work on NaH it was suggested that pressure shifts present for the infrared measurements might account for the difficulty. Initially it seemed that explanation might apply to LiH (and LiD) for which pressures of 25 Torr of H_2 affected the infrared measurements made in this laboratory. However, when only the high-pressure measurements were included, the fit to a potential function was still unsatisfactory and the deviations were quite systematic (showing a strong J-dependence). On the other hand the data could be fit quite well to empirical rovibrational constants. It should be emphasized that this discussion applies for all four isotopic species of LiH treated as though they were four separate molecules (i.e., four separate fits were made with similar results).

In the case of the analysis of the $^{12}C^{16}O$ data, the potential function fit was unable to reproduce the heterodyne measurements of the high-J rotational transitions (*23*). The deviations were systematic and as large as 3 MHz whereas the experimental accuracy was better than 0.1 MHz. In an unpublished analysis of the heterodyne measurements on HF (*24*), we found similar problems. In all these cases a fit to empirical rovibrational constants was quite satisfactory.

The most likely causes of such problems are (a) experimental error or experimental artifacts (such as pressure shifts), (b) errors in the computer program used in the analysis, or (c) error in the model used in the calculation.

It seems unreasonable to invoke experimental artifacts to explain all the observed deviations. Many of the heterodyne measurements were quite independent measurements and some had been independently repeated. One of the reasons we made the far-infrared rotational spectrum measurements on LiH and LiD was to verify that error in the near-infrared frequency measurements was not the cause of the problem.

The computer program used in the fit was subject to a number of different types of tests and the results always agreed with other results. For heavy molecules the potential function fit always agreed with the fit of the rovibrational constants.

Model errors can be divided into two categories, either inadequate representation of the model or else an inadequate model. Bouanich and co-workers (*18*, *19*), Ogilvie and Tipping (*20*), and Bryukhanov *et al.* (*25*) have independently verified Dunham's original work and have added higher-order terms to his expressions. The computer program used in the potential function fit incorporates the higher-order terms so we believe that the difficulty does not arise from any truncation of the Dunham expressions or any other inadequacy in the representation or application of Dunham's model.

Since we are dealing with energy levels that are in all cases down near the bottom of the potential well, the limitation of the Dunham potential to $|(r - r_e)/r_e| < 1$ is not a problem.

We believe the problem lies in the Born–Oppenheimer approximation which results in ignoring the effects of coupling to the other electronic states. This has been dealt

with to some extent by Watson (*13*) who showed how it might affect the isotopic dependence of the rovibratoinal constants but without describing any specific changes that might be made in the Dunham potential function.

Different energy states are often coupled through their angular momentum. In Dunham's potential energy expressions the same potential function is applied for all values of the rotational angular momentum, whereas we expect that the greater the rotational angular momentum the stronger will be the coupling with certain other electronic states. This means that the potential function for high J values will not be the same as the rotationless potential function because the mixing with other electronic states will depend on the rotational angular momentum. The difference will be very small, but we are now dealing with differences on the order of 1.0 MHz out of 4 THz, or about 3 parts in 10^7, in the case of CO, and about one part in 10^5 in the case of LiH.

In conclusion, we find that even when each isotopic species is fit separately the Dunham potential function is unable to fit the observed transitions to within experimental error and the deviations are systematic in terms of increasing J values. This behavior has also been noted for NaH, CO, and HF. This is interpreted as due to the failure of the Born–Oppenheimer approximation. What is needed is a potential function that will take into account the J-dependent mixing of other electronic states with the ground electronic state of LiH and LiD. Until such a potential function is developed, it will be necessary to fit accurate measurements of light molecules to empirical rovibrational constants without regard to any model potential function.

Even with an imperfect fit to the Dunham potential function, fairly accurate values can be determined for the first three or four terms in the Dunham potential function. Since the higher-order rovibrational constants such as Y_{13}, Y_{04}, Y_{14}, Y_{05}, and Y_{06} depend on those lower-order terms in the Dunham potential, they can still be estimated quite reliably. For most purposes one or two digit accuracy is sufficient. It is the terms that depend on a_i where $i > 4$, that are poorly determined by fitting the data to a Dunham potential function.

For heavier molecules, such as KF, the Dunham potential function seems adequate to fit measurements with the accuracy currently available.

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